

IT-SIMPLERA INSTITUTE

Network Administration Internship Program

WEEK 02 ASSIGNMENT REPORT

IP Addressing (FLSM & VLSM) and VLAN Configuration

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1. Objectives

- Understand and apply Fixed Length Subnet Masking (FLSM) for equal-size subnetting.
- Understand and apply Variable Length Subnet Masking (VLSM) for efficient, need-based IP allocation.
- Calculate network address, subnet mask, broadcast address, and usable host range for each subnet.
- Understand the purpose and advantages of VLANs in enterprise networks.
- Configure and verify VLANs, access ports, and trunk links using Cisco Packet Tracer and GNS3.

2. Tools Required and Setup

2.1 Cisco Packet Tracer

What it is: a free Cisco network simulator for building and testing topologies.

- Download: requires a free Cisco Networking Academy (NetAcad) account — sign up at netacad.com, enroll in the free "Getting Started with Cisco Packet Tracer" course, then download Packet Tracer from the course's Resources tab.
- Install and open it; you get a blank workspace with device categories on the bottom-left (Routers, Switches, End Devices, Connections).
- Used for: building the VLAN topology, configuring switches via CLI, and generating the .pkt file required for submission.

2.2 GNS3

What it is: a free, open-source network emulator that runs real Cisco IOS images for higher-fidelity labs.

- Download: gns3.com/software (GNS3 + GNS3 VM). Free account required.
- GNS3 does not ship Cisco IOS images (licensing) — you need a Layer-2 capable image, e.g. an IOSvL2 switch image, imported via GNS3 > Edit > Preferences > IOS on UNIX/IOSv, or use GNS3's built-in generic "Ethernet Switch" node for basic VLAN tagging if you don't have an IOSvL2 image.
- Used for: recreating the same VLAN topology to demonstrate the concept works across two different tools, as required by the task.

2.3 PuTTY

- What it is: a free SSH/Telnet/console client for Windows.
- Download: putty.org.
- Used for: connecting to GNS3 device consoles over Telnet (GNS3 auto-assigns a console port per device — right-click the device > Console, or open PuTTY manually with Connection type: Telnet, Host: 127.0.0.1, Port: as shown in GNS3).

2.4 Calculator / Worksheet

- A subnet calculator (e.g. subnet-calculator.com) is useful to double-check the manual FLSM/VLSM math shown in Sections 3 and 4 — but you should show the manual calculation steps in your worksheet, not just the tool output, since the interview evaluates your understanding of the math.

3. FLSM (Fixed Length Subnet Masking)

FLSM divides a network into equal-size subnets. Every subnet gets the same number of host addresses, regardless of how many hosts it actually needs — simple to plan, but wastes addresses when department sizes differ.

Method

- Determine how many subnets are needed.
- Find the smallest power of 2 $\geq$ required subnets (this tells you how many bits to borrow from the host portion).
- New prefix = original prefix + borrowed bits.
- Subnet increment = $256 \div 2^{(\text{borrowed bits})}$ applied at the octet where borrowing occurs.
- List each subnet: Network address, Usable host range, Broadcast address.

Problem 1: 192.168.1.0/24 → 4 equal subnets

Bits borrowed: 2 | New prefix: /26 | Subnet mask: 255.255.255.192 | Increment: 64

- 192.168.1.0/26 Host: .1–.62 Bcast: .63
- 192.168.1.64/26 Host: .65–.126 Bcast: .127
- 192.168.1.128/26 Host: .129–.190 Bcast: .191
- 192.168.1.192/26 Host: .193–.254 Bcast: .255

Problem 2: 192.168.2.0/24 → 8 equal subnets

Bits borrowed: 3 | New prefix: /27 | Subnet mask: 255.255.255.224 | Increment: 32

- 192.168.2.0/27 (.1–.30, bc .31)
- 192.168.2.32/27 (.33–.62, bc .63)
- 192.168.2.64/27 (.65–.94, bc .95)
- 192.168.2.96/27 (.97–.126, bc .127)
- 192.168.2.128/27 (.129–.158, bc .159)
- 192.168.2.160/27 (.161–.190, bc .191)
- 192.168.2.192/27 (.193–.222, bc .223)
- 192.168.2.224/27 (.225–.254, bc .255)

Problem 3: 10.0.0.0/8 → 16 equal subnets

Bits borrowed: 4 | New prefix: /12 | Subnet mask: 255.240.0.0 | Increment: 16

- 10.0.0.0/12
- 10.16.0.0/12
- 10.32.0.0/12
- 10.48.0.0/12
- 10.64.0.0/12
- 10.80.0.0/12
- 10.96.0.0/12
- 10.112.0.0/12
- ... increments of 16 up to 10.240.0.0/12 (16 total)

Problem 4: 172.16.0.0/16 → 4 equal subnets

Bits borrowed: 2 | New prefix: /18 | Subnet mask: 255.255.192.0 | Increment: 64

- 172.16.0.0/18 (172.16.0.1–172.16.63.254)
- 172.16.64.0/18 (172.16.64.1–172.16.127.254)
- 172.16.128.0/18 (172.16.128.1–172.16.191.254)
- 172.16.192.0/18 (172.16.192.1–172.16.255.254)

Problem 5: 192.168.100.0/24 → 2 equal subnets

Bits borrowed: 1 | New prefix: /25 | Subnet mask: 255.255.255.128 | Increment: 128

- 192.168.100.0/25 (.1–.126, bc .127)
- 192.168.100.128/25 (.129–.254, bc .255)

Problem 6: 192.168.50.0/24 → 16 equal subnets

Bits borrowed: 4 | New prefix: /28 | Subnet mask: 255.255.255.240 | Increment: 16

- 192.168.50.0/28 ... 192.168.50.240/28
- (16 blocks, each 14 usable hosts, increment of 16: .0,.16,.32,.48,.64,.80,.96,.112,.128,.144,.160,.176,.192,.208,.224,.240)

Problem 7: 192.168.20.0/24 → 6 subnets (use 8-block size)

Bits borrowed: 3 | New prefix: /27 | Subnet mask: 255.255.255.224 | Increment: 32

- 192.168.20.0/27
- 192.168.20.32/27
- 192.168.20.64/27
- 192.168.20.96/27
- 192.168.20.128/27
- 192.168.20.160/27 (2 blocks .192/27 and .224/27 left unused/reserved)

Problem 8: 172.20.0.0/16 → 8 equal subnets

Bits borrowed: 3 | New prefix: /19 | Subnet mask: 255.255.224.0 | Increment: 32

- 172.20.0.0/19
- 172.20.32.0/19
- 172.20.64.0/19
- 172.20.96.0/19
- 172.20.128.0/19
- 172.20.160.0/19
- 172.20.192.0/19
- 172.20.224.0/19

Problem 9: 192.168.5.0/24 → 32 equal subnets

Bits borrowed: 5 | New prefix: /29 | Subnet mask: 255.255.255.248 | Increment: 8

- 192.168.5.0/29 ... 192.168.5.248/29
- (32 blocks of 8 addresses / 6 usable hosts each, increment of 8)

Problem 10: 10.10.0.0/16 → 4 equal subnets

Bits borrowed: 2 | New prefix: /18 | Subnet mask: 255.255.192.0 | Increment: 64

- 10.10.0.0/18
- 10.10.64.0/18

- 10.10.128.0/18
- 10.10.192.0/18

4. VLSM (Variable Length Subnet Masking)

VLSM allocates subnets of different sizes matched to actual host requirements, minimizing wasted addresses. The rule of thumb: sort requirements from largest to smallest, allocate the largest block first, then continue allocating from the next available address.

Method

- List all host requirements and sort in descending order.
- For each requirement, find the smallest block size (2^n) that covers hosts + 2 (network + broadcast).
- Assign the block starting at the next free address; usable range = network+1 to broadcast-1.
- Move to the next free address after the broadcast address and repeat for the next requirement.

Scenario 1 — Single Office LAN

Base Network: 192.168.1.0/24 | Departments: Sales (50 hosts), HR (25 hosts), IT (10 hosts), Guest Wi-Fi (5 hosts)

Segment	Hosts Needed	Block/Mask	Network Address	Usable Range	Broadcast
Sales	50	/26 (62 usable)	192.168.1.0/26	192.168.1.1–.62	192.168.1.63
HR	25	/27 (30 usable)	192.168.1.64/27	192.168.1.65–.94	192.168.1.95
IT	10	/28 (14 usable)	192.168.1.96/28	192.168.1.97–.110	192.168.1.111
Guest Wi-Fi	5	/29 (6 usable)	192.168.1.112/29	192.168.1.113–.118	192.168.1.119

Scenario 2 — Corporate Network

Base Network: 172.16.0.0/16 | Engineering (500), Marketing (200), Finance (100), Support (50)

Segment	Hosts Needed	Block/Mask	Network Address	Usable Range	Broadcast
Engineering	500	/23 (510 usable)	172.16.0.0/23	172.16.0.1–172.16.1.254	172.16.1.255
Marketing	200	/24 (254 usable)	172.16.2.0/24	172.16.2.1–.254	172.16.2.255
Finance	100	/25 (126 usable)	172.16.3.0/25	172.16.3.1–.126	172.16.3.127
Support	50	/26 (62 usable)	172.16.3.128/26	172.16.3.129–.190	172.16.3.191

Scenario 3 — Multi-Branch WAN

Base Network: 10.0.0.0/8 | HQ (1000), Branch-1 (500), Branch-2 (200), Branch-3 (100), Branch-4 (50)

Segment	Hosts Needed	Block/Mask	Network Address	Usable Range	Broadcast
HQ	1000	/22 (1022 usable)	10.0.0.0/22	10.0.0.1–10.0.3.254	10.0.3.255
Branch-1	500	/23 (510 usable)	10.0.4.0/23	10.0.4.1–10.0.5.254	10.0.5.255
Branch-2	200	/24 (254 usable)	10.0.6.0/24	10.0.6.1–.254	10.0.6.255
Branch-3	100	/25 (126 usable)	10.0.7.0/25	10.0.7.1–.126	10.0.7.127
Branch-4	50	/26 (62 usable)	10.0.7.128/26	10.0.7.129–.190	10.0.7.191

Scenario 4 — VLAN-Based Addressing (used for the VLAN lab in this report)

Base Network: 192.168.10.0/24 | VLAN 10 Sales (60), VLAN 20 HR (30), VLAN 30 IT (14), Router link Mgmt (2, point-to-point)

Segment	Hosts Needed	Block/Mask	Network Address	Usable Range	Broadcast
VLAN10-Sales	60	/26 (62 usable)	192.168.10.0/26	192.168.10.1–.62	192.168.10.63
VLAN20-HR	30	/27 (30 usable)	192.168.10.64/27	192.168.10.65–.94	192.168.10.95
VLAN30-IT	14	/28 (14 usable)	192.168.10.96/28	192.168.10.97–.110	192.168.10.111
Mgmt Link	2	/30 (2 usable)	192.168.10.112/30	192.168.10.113–.114	192.168.10.115

Scenario 5 — Small Office VLSM

Base Network: 192.168.100.0/24 | Admin (20), Finance (12), Servers (6), Guest Wi-Fi (28)

Segment	Hosts Needed	Block/Mask	Network Address	Usable Range	Broadcast
Guest Wi-Fi	28	/27 (30 usable)	192.168.100.0/27	192.168.100.1–.30	192.168.100.31
Admin	20	/27 (30 usable)	192.168.100.32/27	192.168.100.33–.62	192.168.100.63
Finance	12	/28 (14 usable)	192.168.100.64/28	192.168.100.65–.78	192.168.100.79
Servers	6	/29 (6 usable)	192.168.100.80/29	192.168.100.81–.86	192.168.100.87

5. VLAN Design and Topology

VLANs (Virtual LANs) logically segment a single physical switch/network into separate broadcast domains, improving security (isolating Sales traffic from HR traffic), performance (smaller broadcast domains), and manageability (grouping users by department instead of physical location).

5.1 Topology

- Switch0 (2960) — Fa0/1 — PC0 (VLAN 10 – Sales)
- Switch0 (2960) — Fa0/2 — PC1 (VLAN 20 – HR)
- Switch0 Gi0/1 — TRUNK LINK — Switch1 Gi0/1
- Switch1 (2960) — Fa0/1 — PC2 (VLAN 10 – Sales)
- Switch1 (2960) — Fa0/2 — PC3 (VLAN 20 – HR)

5.2 IP Addressing (from VLSM Scenario 4)

Device	VLAN	IP Address	Subnet Mask	Default Gateway
PC0	10 (Sales)	192.168.10.2	255.255.255.192	192.168.10.1
PC2	10 (Sales)	192.168.10.3	255.255.255.192	192.168.10.1
PC1	20 (HR)	192.168.10.66	255.255.255.224	192.168.10.65
PC3	20 (HR)	192.168.10.67	255.255.255.224	192.168.10.65

Note: gateway addresses (.1 / .65) are only needed if a router-on-a-stick is added for inter-VLAN routing (optional extension — see 5.6).

5.3 Step-by-Step: Cisco Packet Tracer

1. Open Packet Tracer → drag two 2960 switches (Switch0, Switch1) and four generic PCs onto the canvas.
2. Cabling: use Copper Straight-Through for PC–Switch links; Packet Tracer auto-negotiates the Switch–Switch link, or use Copper Straight-Through there too since 2960 ports auto-MDIX.
 - PC0 → Switch0 Fa0/1, PC1 → Switch0 Fa0/2
 - PC2 → Switch1 Fa0/1, PC3 → Switch1 Fa0/2
 - Switch0 Gi0/1 → Switch1 Gi0/1
3. Set static IPs on each PC: click PC → Desktop tab → IP Configuration → enter IP/Subnet Mask from the table in 5.2.
4. Click Switch0 → CLI tab and enter the configuration in 5.4.
5. Repeat the mirrored configuration on Switch1 (5.5).
6. Verify with the commands in Section 6, then test connectivity from PC0's Command Prompt using ping.
7. Save the file as Report_Week2.pkt for submission.

5.4 Switch0 Configuration

```
enable
configure terminal
hostname Switch0
vlan 10
  name Sales
vlan 20
  name HR
exit
interface fastEthernet 0/1
  switchport mode access
  switchport access vlan 10
```

```

no shutdown
exit
interface fastEthernet 0/2
  switchport mode access
  switchport access vlan 20
  no shutdown
exit
interface gigabitEthernet 0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
exit
end
write memory

```

5.5 Switch1 Configuration

```

enable
configure terminal
hostname Switch1
vlan 10
  name Sales
vlan 20
  name HR
exit
interface fastEthernet 0/1
  switchport mode access
  switchport access vlan 10
  no shutdown
exit
interface fastEthernet 0/2
  switchport mode access
  switchport access vlan 20
  no shutdown
exit
interface gigabitEthernet 0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
exit
end
write memory

```

5.6 Optional Extension: Inter-VLAN Routing (Router-on-a-Stick)

By design, VLAN10 and VLAN20 cannot reach each other yet — that is expected and demonstrates segmentation. If your task/interview asks for inter-VLAN connectivity, add a router with a sub-interface per VLAN:

```

! On a router connected to Switch0 Gi0/2 (trunk)
interface gigabitEthernet 0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.192
interface gigabitEthernet 0/0.20
  encapsulation dot1Q 20
  ip address 192.168.10.65 255.255.255.224
interface gigabitEthernet 0/0
  no shutdown

```

Remember to also add "switchport trunk allowed vlan 10,20" on the switch port facing the router, and set PC default gateways to .1 (VLAN10) / .65 (VLAN20).

5.7 Step-by-Step: GNS3

- 1. Open GNS3 → confirm a Layer-2 capable node is available (IOSvL2 image imported under Edit > Preferences > IOS on UNIX/IOSv, or the built-in "Ethernet Switch" template for basic tagging).
- 2. Drag two switch nodes onto the canvas and rename them Switch0 / Switch1 to mirror the Packet Tracer topology.
- 3. Add four VPCS (Virtual PC) nodes for PC0–PC3 and connect them exactly as in 5.1.
- 4. Start all nodes (green play button), right-click each switch → Console — this opens a Telnet session (PuTTY connects automatically, or manually via PuTTY: Telnet, 127.0.0.1, port shown in GNS3).
- 5. Apply the same CLI configuration as 5.4/5.5 on the respective switch consoles.
- 6. On each VPCS node, set the IP: open its console and run:

```
ip 192.168.10.2/26 192.168.10.1 ; example for PC0
ip 192.168.10.66/27 192.168.10.65 ; example for PC1
```

- 7. Save the project (File > Save project as...) — this produces the GNS3 project folder required for submission.

6. Verification Commands and Expected Evidence

Take a screenshot of each command's output for your Screenshots folder.

show vlan brief

Confirms VLAN10 and VLAN20 exist and lists which access ports belong to each — Fa0/1 should show under VLAN10, Fa0/2 under VLAN20.

show interfaces trunk

Confirms Gi0/1 is in trunking mode and that VLANs 10 and 20 are in the "allowed" and "active" VLAN list.

Ping tests

- PC0 (VLAN10) → PC2 (VLAN10): should SUCCEED — same VLAN, same broadcast domain.
- PC1 (VLAN20) → PC3 (VLAN20): should SUCCEED — same VLAN.
- PC0 (VLAN10) → PC1 (VLAN20): should FAIL without a router — proves VLAN isolation (this is expected, correct behavior, not a fault).
- If Section 5.6 (router-on-a-stick) is added, PC0 → PC1 should then SUCCEED, proving inter-VLAN routing works.

7. Challenges Faced and Solutions

(Fill in with your actual experience before submission — examples below.)

Challenge	Solution
Trunk link not passing traffic between switches	Confirmed both ends were set to switchport mode trunk and that the VLANs were included in the trunk-allowed list.
PC could not ping across VLANs	Confirmed this was expected behavior since no router was configured; documented as intended VLAN isolation.
GNS3 switch had no VLAN CLI support	Used the IOSvL2 image instead of the generic Ethernet Switch node to get full Cisco IOS VLAN commands.

8. Learning Outcomes

- Learned to calculate subnet masks, network/broadcast addresses, and usable host ranges for both FLSM and VLSM.
- Understood when FLSM (simplicity) is preferable versus VLSM (address efficiency).
- Gained hands-on experience configuring VLANs, access ports, and trunk links on Cisco switches.
- Learned to verify configurations using show vlan brief, show interfaces trunk, and ping tests.
- Understood why VLANs improve security and performance by segmenting broadcast domains.

8.5 Screenshots — GNS3 Implementation Evidence

The topology below and the following CLI/ping outputs are the actual verification evidence captured from the GNS3 lab for this task.

Final Topology (GNS3)

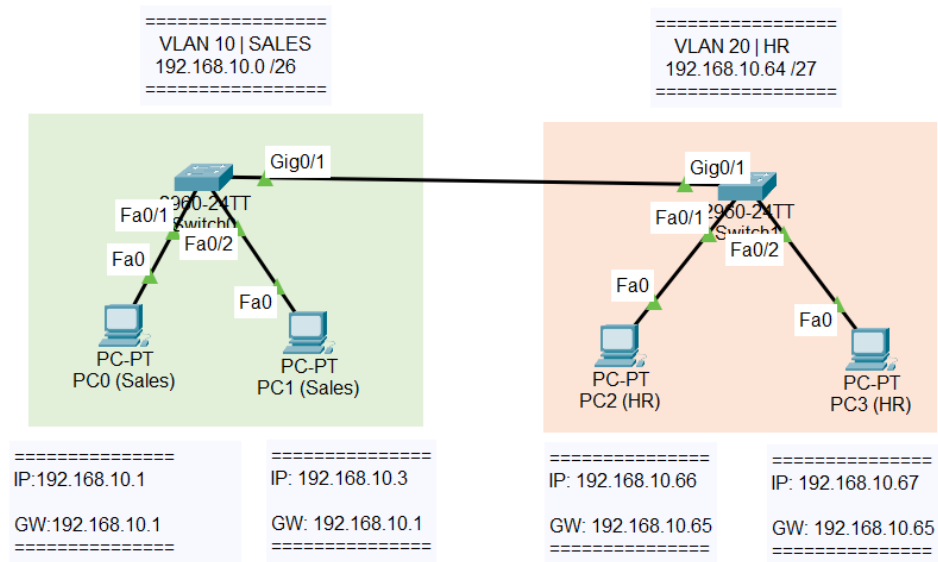
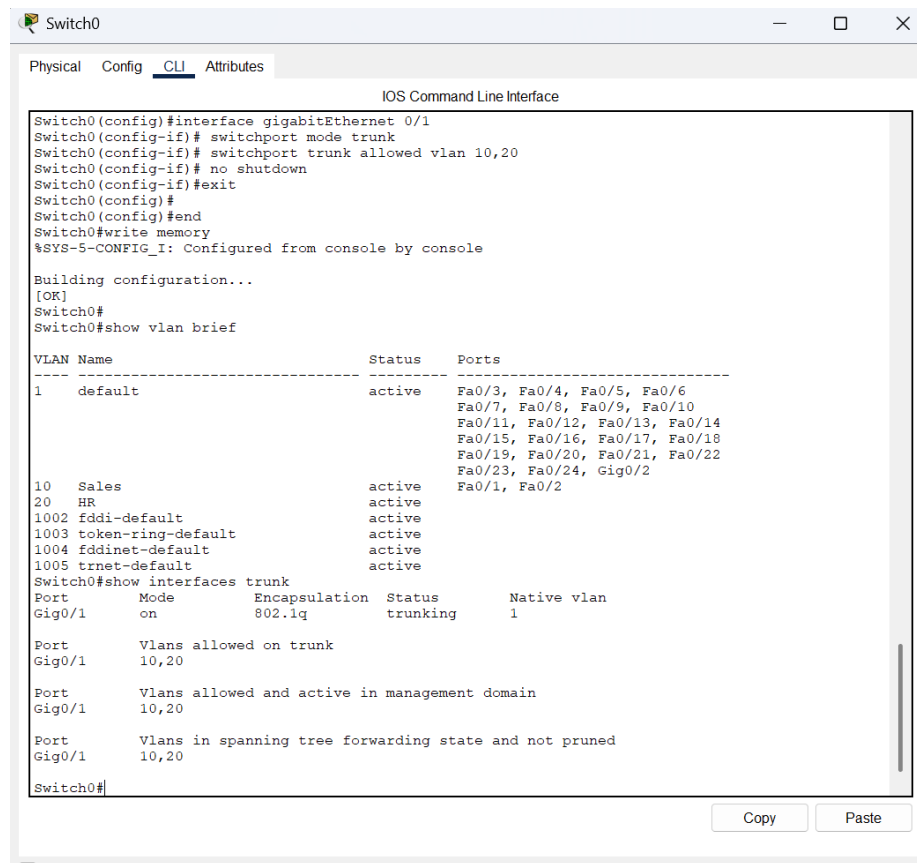


Figure 1: GNS3 topology - Switch1 (VLAN10 Sales, VLAN20 HR) trunked to Switch2 (VLAN10 Sales, VLAN20 HR) via Gig0/1

Switch0 - VLAN and Trunk Configuration



The screenshot shows a terminal window titled 'Switch0' with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the following commands and output:

```
Switch0(config)#interface gigabitEthernet 0/1
Switch0(config-if)# switchport mode trunk
Switch0(config-if)# switchport trunk allowed vlan 10,20
Switch0(config-if)# no shutdown
Switch0(config-if)#exit
Switch0(config)#
Switch0(config)#end
Switch0#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch0#
Switch0#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10 Sales	active	Fa0/1, Fa0/2
20 HR	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch0#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    10,20

Port      Vlans allowed and active in management domain
Gig0/1    10,20

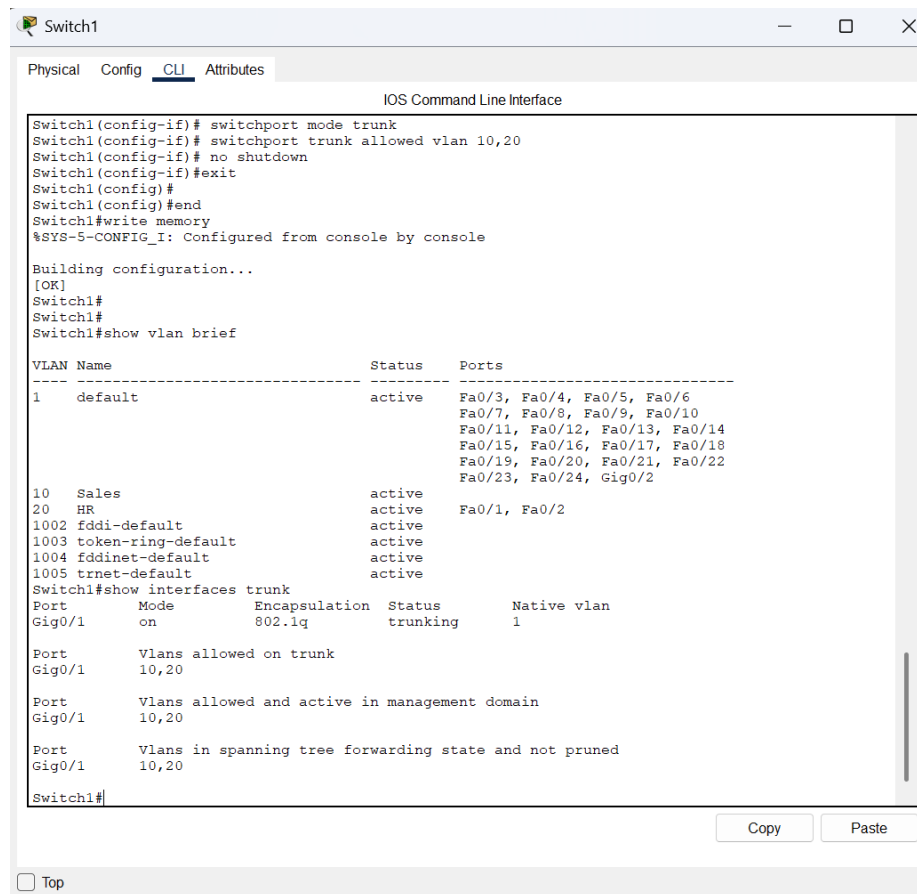
Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    10,20

Switch0#
```

At the bottom of the terminal window, there are 'Copy' and 'Paste' buttons.

Figure 2: Switch0 CLI - trunk configured on Gig0/1; show vlan brief confirms VLAN10 (Sales) and VLAN20 (HR); show interfaces trunk confirms VLANs 10,20 allowed and active

Switch1 - VLAN and Trunk Configuration



The screenshot shows the Switch1 CLI interface with the following content:

```
Switch1
Physical Config CLI Attributes
IOS Command Line Interface

Switch1(config-if)# switchport mode trunk
Switch1(config-if)# switchport trunk allowed vlan 10,20
Switch1(config-if)# no shutdown
Switch1(config-if)# exit
Switch1(config)#
Switch1(config)# end
Switch1# write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch1#
Switch1#
Switch1# show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10 Sales	active	
20 HR	active	Fa0/1, Fa0/2
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch1# show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1

```
Switch1#
```

Additional output for Gig0/1:

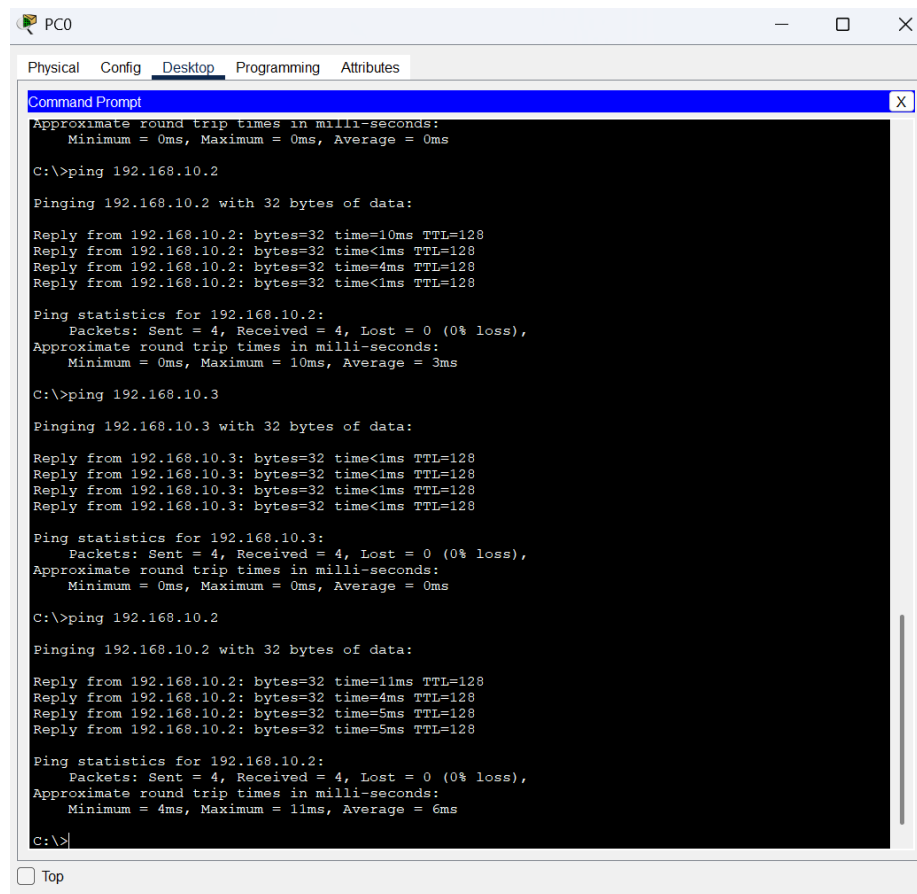
```
Port Gig0/1 Vlan allowed on trunk 10,20
Port Gig0/1 Vlan allowed and active in management domain 10,20
Port Gig0/1 Vlan in spanning tree forwarding state and not pruned 10,20
```

Buttons: Copy, Paste

Top

Figure 3: Switch1 CLI - matching trunk configuration on Gig0/1; show vlan brief and show interfaces trunk confirm identical VLAN setup

Ping Verification - PC0 (VLAN10)



The screenshot shows a PC0 window with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the results of three ping commands: a preliminary ping to 192.168.10.2, a ping to 192.168.10.3, and a final ping to 192.168.10.2. All pings are successful with 0% loss.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=10ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=4ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

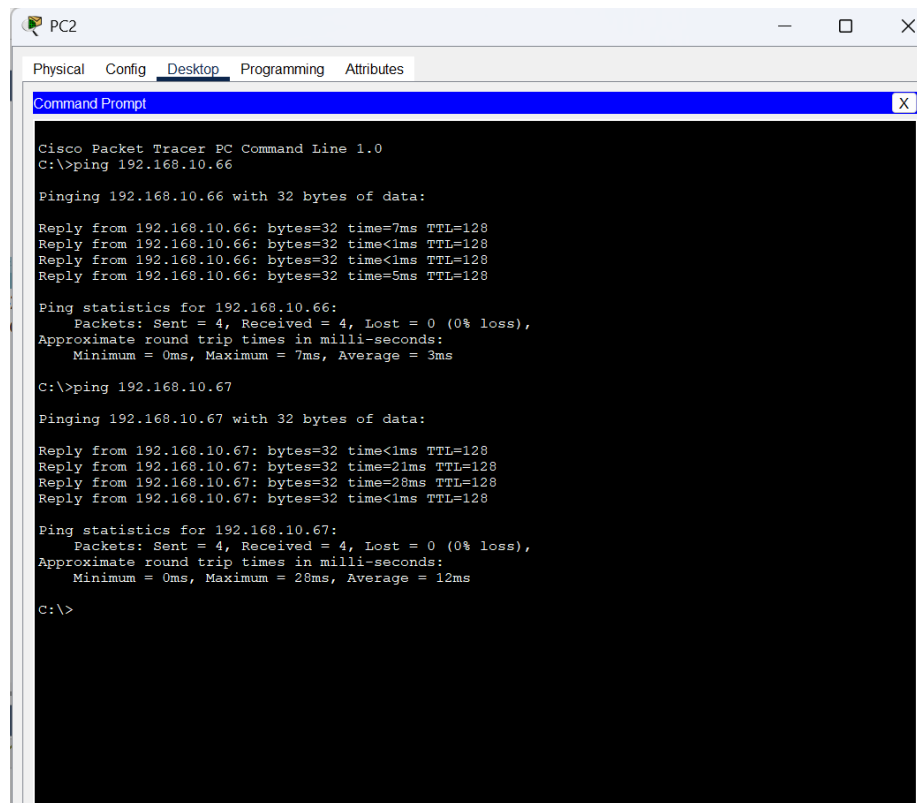
Reply from 192.168.10.2: bytes=32 time=11ms TTL=128
Reply from 192.168.10.2: bytes=32 time=4ms TTL=128
Reply from 192.168.10.2: bytes=32 time=5ms TTL=128
Reply from 192.168.10.2: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 11ms, Average = 6ms

C:\>
```

Figure 4: PC0 successfully pings 192.168.10.2 and 192.168.10.3 - confirms VLAN10 connectivity across the trunk link (0% loss)

Ping Verification - PC2 (VLAN20)



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC2. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt shows the following output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.66

Pinging 192.168.10.66 with 32 bytes of data:

Reply from 192.168.10.66: bytes=32 time=7ms TTL=128
Reply from 192.168.10.66: bytes=32 time<1ms TTL=128
Reply from 192.168.10.66: bytes=32 time<1ms TTL=128
Reply from 192.168.10.66: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.10.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>ping 192.168.10.67

Pinging 192.168.10.67 with 32 bytes of data:

Reply from 192.168.10.67: bytes=32 time<1ms TTL=128
Reply from 192.168.10.67: bytes=32 time=21ms TTL=128
Reply from 192.168.10.67: bytes=32 time=28ms TTL=128
Reply from 192.168.10.67: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.67:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 28ms, Average = 12ms

C:\>
```

Figure 5: PC2 successfully pings 192.168.10.66 and 192.168.10.67 - confirms VLAN20 connectivity across the trunk link (0% loss)

Result: same-VLAN pings across the trunk succeeded with 0% packet loss in every test, confirming both correct VLAN tagging on the trunk link and correct access-port assignment on each switch.

9. Conclusion

This week's task strengthened both the theoretical and practical sides of network segmentation: FLSM and VLSM calculations for structured IP address planning, and VLAN configuration in Packet Tracer and GNS3 for enforcing that segmentation at Layer 2. The verification evidence confirms that VLANs correctly isolate traffic by department while trunk links carry tagged traffic between switches.